

Siamese Network

Motivation:



↙ In which cases Classification
Can fail?

when
you
see an
eg.
which is
not in
training
set.

or

for intelligent vision $\Rightarrow$ Comparison
do we have
something which
is better than
classification

or

↙ Do humans always
Not necessarily do classification

→ Humans also do
Comparison

Also helps in
evaluating
differences.

Comparison

- 1) Comparison
 - 2) Evaluate Differences between two samples
- Defining a new loss function.

Application $\Rightarrow$ 'face recognition'

- Every new member adds a class
- For classification you need many samples for the same class

"New loss function"

vector Algebra $\Rightarrow$

$$d(x, y) = \sqrt{\sum_{i=1}^n (x_i - y_i)^2}$$

$$x = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}$$

$$y = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix}$$

Euclidean distance is a good metric to quantify difference between two vectors.

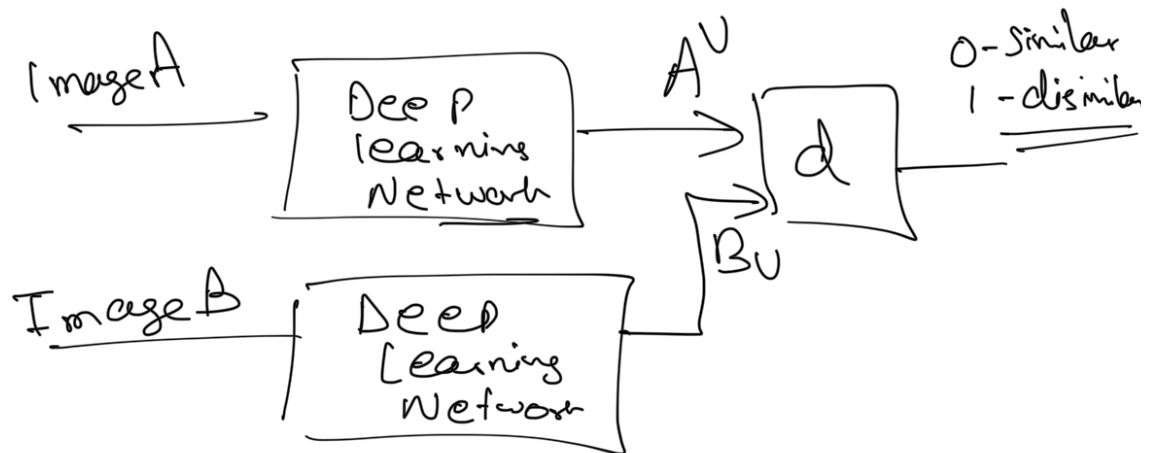
Let A and B be two images

$$d(\vec{A}, \vec{B}) = \|\vec{A} - \vec{B}\| \Rightarrow \text{Norm}$$

distance
between
two
images

$A^U, B^U \Rightarrow$ vectors
↓
"Embedding vectors"

Representation of the
images in form
of vectors as
"perceived by
Siamese Network the network"



Two branches have the

- 1) Same Network
- 2) Share the same weights

Loss function for Siamese Network

- Contrastive Loss
- Triplet Loss

x_1, x_2 Embed vectors from two branches

$y = 0$: Similar pair

$y = 1$: Dissimilar pair

$$d(x_1, x_2) = \|x_1 - x_2\|_2 = \sqrt{\sum_{i=1}^n (x_{1i} - x_{2i})^2}$$

Contrastive loss:

$$\mathcal{L}(\underbrace{y}_{\substack{\text{true} \\ \text{class}}}, \underbrace{d}_{d(x_1, x_2)}) = \frac{1}{2} (1-y) d^2 + \frac{1}{2} y (\max(0, m-d))^2$$

$m \Rightarrow$ hyperparameter
(margin)

During Training

$y = 0$ (similar input)

$$\mathcal{L}(y, d) = \frac{1}{2} \underline{d^2}$$

weights are obtained
so that $\mathcal{L}(y, d)$ is
minimized

$\downarrow$
 d is minimized.

$\downarrow$
 $d(A^u, B^u)$ is minimized

$y = 1$ (dissimilar inputs)

$$\mathcal{L}(y, d) = \frac{1}{2} y \max(0, \underline{m-d})^2$$

$m \Rightarrow$ margin
decides that two
dissimilar images
should at least
be this much
apart.

Training set

(A, B) — 0

— 1

$$L(y, d) = \begin{cases} 0 & d > m \\ \frac{1}{2}(m-d)^2 & d < m \end{cases} \Rightarrow \begin{cases} \text{no learning} \\ \text{required} \\ \text{no weight} \\ \text{update} \\ \text{required} \end{cases}$$

Triplet Loss:

Three Image {

- Anchor (A) $\Rightarrow$ reference image
- Positive (P) $\Rightarrow$ positive class
(P and A are similar)
- Negative (N) $\Rightarrow$ Negative class
(P and N are dissimilar)

(A, P, N)

$\mathcal{L}(A, P) = \|f(A) - f(P)\|_2$

$$d(A, P) = \|f(A) - f(P)\|_2$$

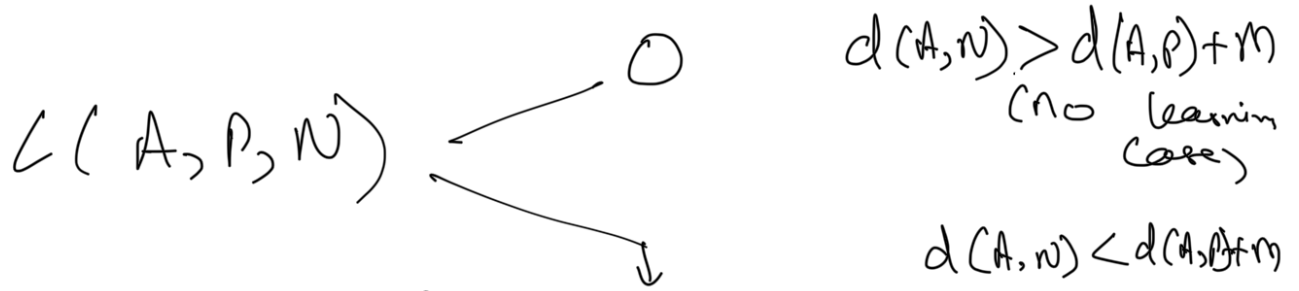
$$= \|A^V - P^V\|_2$$

$$d(A, N) = \|f(A) - f(N)\|_2$$

$$= \|A^V - N^V\|_2$$

$\frac{1}{n} \ln \left(\frac{n!}{n^n} \right) = \frac{1}{n} \ln n! - \ln n$

$\hookrightarrow L(A, P, N) = \max(\emptyset, d(A, P) - d(A, N))$
 triplet loss



$\min (d(A, P) - d(A, N) + m)$

find the weights such that this is minimized.